Auteur: Michael Livchits
Studierichting: Informatica
Oefeningenreeks 3A nr. 4

$$f(x) = \frac{1}{x^2}$$

Berekent men de afgeleide functie als volgt:

$$f(x+h) = \frac{1}{(x+h)^2}$$

$$\Rightarrow f(x+h) - f(x) = \frac{1}{(x+h)^2} - \frac{1}{x^2} = \frac{-2xh - h^2}{(x+h)^2 x^2}$$

$$\Rightarrow \frac{f(x+h) - f(x)}{h} = \frac{-2xh - h^2}{(x+h)^2 x^2 h} = \frac{-2x - h}{(x+h)^2 x^2}$$

$$\Rightarrow f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \to 0} \frac{-2x - h}{(x+h)^2 x^2} = \frac{-2}{x^3}$$

Nu met nx^{n-1}

$$f(x) = \frac{1}{x^2} = x^{-2}$$

$$f'(x) = -2x^{-3} = \frac{-2}{x^3}$$

References